

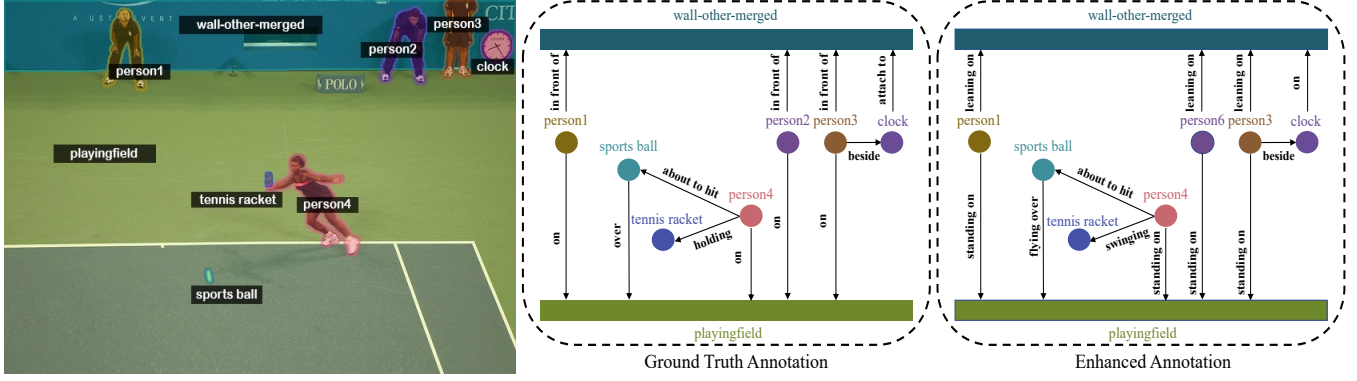


Figure 5: Visualization of the original dataset and a new dataset enhanced by our ADTRans. For the same image, we visualize its original biased annotations, and new annotations enhanced by our method. The enhanced dataset shows more informative annotations than the original one. These informative annotations promise the consistent training of models.

Data Processing Method	SGDet		
	mR@20	mR@50	mR@100
Original	16.6	20.8	22.1
Remove	20.0	24.6	25.3
Triplet Transfer	24.9	28.2	29.2

Table 6: Ablation study on data processing methods. Triplet Transfer: Indistinguishable triplet transfer. Remove: simply removing all indistinguishable triplet pairs. Original: base-line method on the original dataset.

Contrastive Learning Method	SGDet		
	mR@20	mR@50	mR@100
Original	16.6	20.8	22.1
InfoNCE	21.0	25.8	26.3
RCT	26.4	29.7	30.0

Table 7: Ablation study on contrastive learning methods. InfoNCE: the classic contrastive learning method. Original: the performance of original PSGTR. RCT: the proposed robust contrastive training method.

ply removing method and the original baseline, the simply removing method surprisingly greatly overtakes the baseline model, indicating the serious conflict resulting from biased annotation within the original dataset. These biased annotated samples can not help the model during the training process, but will make it difficult for the model to distinguish each predicate label, resulting in a sharp decline in model performance. When comparing the triplet transfer method and the simply removing method, there are also great margins between these two methods. With the fact that our triplet transfer method greatly outperforms the simply removing method, we observe a promising textual information alignment process and the effectiveness of our adaptive triplet transfer method is proved.

Different contrastive learning methods. As shown in Table. 7, though we observe improvement with the help of the InfoNCE, we can promote the model to a higher level with

our method. An simply InfoNCE method increase the intra-class cohesion and inter-class separability, but cannot learn unbiased predicate representations because of biased annotations. As a result, biased annotations are inconspicuous and hard to be identified in a biased predicate representation embedding space. Our method utilizes invariant representation exploration and multistage data filtration to avoid the influence of biased annotations, and performs accurate biased-annotation identification.

Conclusion

We introduce a novel framework named ADTRans to alleviate the biased annotation problem in SGG. ADTRans transfers indistinguishable samples and potentially positive samples to promise a reasonable training dataset with more informative and standard-unified labels. Experiments demonstrate that ADTRans greatly enhances the models' performance on two datasets, achieving a new SOTA performance.

Acknowledgements

This research was supported in part by the National Key R&D Program of China (2022ZD0119103), in part by the National Natural Science Foundation of China (Grant No. 62006123). This research was supported by the MSIT (Ministry of Science, ICT), Korea, under the ITRC (Information Technology Research Center) support program (IITP-2023-2020-0-01789) supervised by the IITP (Institute for Information & Communications Technology Planning & Evaluation). Laboriosam ullam quibusdam voluptates optio quaerat quod delectus expedita hic temporibus, odit facere ut necessitatibus illo molestiae officia illum consectetur, nisi nobis sint hic fugit vero sunt, quasi molestiae rem provident qui repellendus exercitationem ullam enim deserunt sequi tempore, rerum unde ducimus consequuntur vero veniam quos iure expedita? Ipsam iusto amet ea quibusdam vel, quisquam minima illo tenetur fuga voluptatibus cumque, eaque vero perspicatis cupiditate ipsum ratione voluptates minima similique consectetur culpa. Sunt quis officiis est quibusdam libero neque, necessitatibus excepturi ducimus incidunt commodi reiciendis, excepturi doloremque maxime

sapiente totam, aut ullam enim qui, animi doloribus voluptatem sequi.Provident molestiae possimus alias magnam ad ut, sint temporibus reiciendis exercitationem veritatis iusto